Kuang-Da Wang

Ph.D. Candidate, Computer Science and Engineering
National Yang Ming Chiao Tung University

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Education

National Yang Ming Chiao Tung University Sep 2023 – Sep 2027 (expected)
Ph.D. in Computer Science and Engineering Hsinchu, Taiwan
Advisers: Prof. **Wen-Chih Peng** and Prof. **Ping-Chun Hsieh** | GPA: 3.83/4.00

National Yang Ming Chiao Tung University Sep 2021 – Aug 2023
M.S. in Networking Engineering Hsinchu, Taiwan
Adviser: Prof. **Wen-Chih Peng** | GPA: 3.90/4.00

National Cheng Kung University Sep 2018 – Jun 2021
B.S. in Engineering Science Tainan, Taiwan
GPA: 3.89/4.00
Prior study in the Department of Psychology, Sep 2017 – Jun 2018; transferred to Engineering Science.

Research & Industry Experience

Carnegie Mellon University May 2026 – Present
Visiting Researcher, WAVLab, Language Technologies Institute Pittsburgh, PA, USA
Research Adviser: Prof. **Shinji Watanabe** | Visit expected to conclude May 2027

- Developing reward-guided latent-space test-time optimization for multilingual automatic speech recognition, using error localization and reference-free search to adapt frozen speech models.

Sony Group Corporation Aug 2025 – Feb 2026
LLM Research Intern Tokyo, Japan
Mentors: **Zhao Wang** and the Sony research team

- Developed agentic model predictive questioning control for multi-turn visual design, balancing alignment with user interaction cost (**ICML 2026**, co-first author).
- Designed a pipeline for structured webpage collection, multimodal web generation, and section-level evaluation (**KDD 2026**, co-first author).
- Co-developed automated visual-prompt discovery through semantic exploration to address perception failures in vision-language models (**ECCV 2026**, co-author).

National Yang Ming Chiao Tung University Sep 2023 – Present
Ph.D. Researcher Hsinchu, Taiwan
Advanced Database System Laboratory (ADSL) and RL & Bandits Lab (REAL)
Advisers: Prof. **Wen-Chih Peng** and Prof. **Ping-Chun Hsieh**

- Developed textual model predictive control for inference-time LLM alignment (**ICLR 2026**) and book-length machine translation evaluation (**EMNLP 2025**), through industry–academia research collaboration with **NVIDIA Research Taiwan**.
- Developed badminton player imitation and realistic tactical simulators (**ECML PKDD 2024**; **AAAI 2024 demo**; **MIT Sloan 2025 research competition finalist**). Open source: CoachAI+, badmintonGPT.

Taiwan Semiconductor Manufacturing Company Jun 2023 – Dec 2024
AI Intern, AI4BI Hsinchu, Taiwan

- Evaluated agentic and decomposed information retrieval for financial report generation; developed table-extraction applications for image-based financial documents.

Publications

* Equal contribution.

Peer-Reviewed Conference Papers

[C14] ATLAS: Agentic ASR Tuning for Research Labor Scaling.
Masao Someki, Samuele Cornell, **Kuang-Da Wang**, Shinji Watanabe.
Accepted to the IEEE Spoken Language Technology Workshop (SLT 2026); forthcoming.

[C13] CAME: Company-Aware Evidence-Memory Experts for Interpretable Quarter-Ahead Revenue Forecasting.
Ya-Wen Wu, Meng-Fen Chiang, **Kuang-Da Wang**, Wen-Chih Peng.
Accepted to the Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Main Conference; forthcoming.

[C12] Visual Prompt Discovery via Semantic Exploration.
Jaechang Kim, Yotaro Shimose, Zhao Wang, **Kuang-Da Wang**, Jungseul Ok, Shingo Takamatsu.
In Proceedings of the 19th European Conference on Computer Vision (ECCV 2026).

[C11] WebGen-V Bench: Structured Representation for Enhancing Visual Design in LLM-based Web Generation and Evaluation.
Kuang-Da Wang*, Zhao Wang*, Wei-Yao Wang, Yotaro Shimose, Shingo Takamatsu.
In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026).

[C10] Agentic Model Predictive Questioning Control in Visual Design.
Kuang-Da Wang*, Zhao Wang*, Wei-Yao Wang, Yotaro Shimose, Jaechang Kim, Shingo Takamatsu.
In Proceedings of the 43rd International Conference on Machine Learning (ICML 2026).

[C9] Benchmarking Agentic Newswriting via Journalistic Workflows.
Yen-Che Chien*, **Kuang-Da Wang***, Wei-Yao Wang, Wen-Chih Peng.
In Findings of the Association for Computational Linguistics: ACL 2026.

[C8] Test-Time Alignment for Large Language Models via Textual Model Predictive Control.
Kuang-Da Wang*, Teng-Ruei Chen*, Yu Heng Hung, Guo-Xun Ko, Shuoyang Ding, Yueh-Hua Wu, Yu-Chiang Frank Wang, Chao-Han Huck Yang, Wen-Chih Peng, Ping-Chun Hsieh.
In Proceedings of the 14th International Conference on Learning Representations (ICLR 2026).

[C7] Adapting to Evolving Data: Test-Time Expert Aggregation for Imbalanced Tabular Regression.
Yung-Chien Wang, **Kuang-Da Wang**, Wei-Yao Wang, Wen-Chih Peng.
In Proceedings of the 19th ACM International Conference on Web Search and Data Mining (WSDM 2026), Short Paper.

[C6] Extending Automatic Machine Translation Evaluation to Book-Length Documents.
Kuang-Da Wang*, Shuoyang Ding*, Chao-Han Huck Yang, Ping-Chun Hsieh, Wen-Chih Peng, Vitaly Lavrukhin, Boris Ginsburg.
In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2025), Main Conference.

[C5] Template-Based Financial Report Generation in Agentic and Decomposed Information Retrieval.
Yong-En Tian, Yu-Chien Tang, **Kuang-Da Wang**, An-Zi Yen, Wen-Chih Peng.
In Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2025), Short Paper.

[C4] DDOT: A Derivative-Directed Dual-Decoder Ordinary Differential Equation Transformer for Dynamic System Modeling.
Yang Chang, **Kuang-Da Wang**, Ping-Chun Hsieh, Cheng-Kuan Lin, Wen-Chih Peng.
In Proceedings of the 29th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025).

[C3] APAR: Modeling Irregular Target Functions in Tabular Regression via Arithmetic-Aware Pre-Training and Adaptive-Regularized Fine-Tuning.
Hong-Wei Wu, Wei-Yao Wang, **Kuang-Da Wang**, Wen-Chih Peng.
In Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI 2025).

Publications (continued)

[C2] Offline Imitation of Badminton Player Behavior via Experiential Contexts and Brownian Motion.
Kuang-Da Wang, Wei-Yao Wang, Ping-Chun Hsieh, Wen-Chih Peng.
In Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2024), Applied Data Science Track.

[C1] Root Cause Analysis in Microservice Using Neural Granger Causal Discovery.
Cheng-Ming Lin, Ching Chang, Wei-Yao Wang, **Kuang-Da Wang**, Wen-Chih Peng.
In Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI 2024).

Workshops & Special Tracks

[P6] Nvidia-Nemo's WMT 2025 Metrics Shared Task Submission.
Brian Yan, Shuoyang Ding, **Kuang-Da Wang**, Siqi Ouyang, Oleksii Hrinchuk, Vitaly Lavrukhin, Boris Ginsburg.
In Proceedings of the 10th Conference on Machine Translation (WMT 2025), Metrics Shared Task.

[P5] RallyDiffuser: A Representation-Guided Diffusion Model Framework for Strategic Planning in Badminton.
Bing-Zhi Ke, **Kuang-Da Wang**, Wen-Chih Peng.
In Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), Extended Abstract.

[P4] The CoachAI Badminton Environment: Bridging the Gap between a Reinforcement Learning Environment and Real-World Badminton Games.
Kuang-Da Wang, Yu-Tse Chen, Yu-Heng Lin, Wei-Yao Wang, Wen-Chih Peng.
In Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI 2024), Demonstration Track.

[P3] The CoachAI Badminton Environment: A Novel Reinforcement Learning Environment with Realistic Opponents.
Kuang-Da Wang, Wei-Yao Wang, Yu-Tse Chen, Yu-Heng Lin, Wen-Chih Peng.
In Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI 2024), Student Abstract.

[P2] Enhancing Badminton Player Performance via a Closed-Loop AI Approach: Imitation, Simulation, Optimization, and Execution.
Kuang-Da Wang.
In Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM 2023), Doctoral Consortium.

[P1] A Reinforcement Learning Badminton Environment for Simulating Player Tactics.
Li-Chun Huang, Nai-Zen Hsueh, Yen-Che Chien, Wei-Yao Wang, **Kuang-Da Wang**, Wen-Chih Peng.
In Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI 2023), Student Abstract.

Additional Research Outputs

CoachAI+ Badminton Environment: Realistic Badminton Game Simulator for Enhancing Player Performance.
Wen-Chih Peng, **Kuang-Da Wang**, Yen-Che Chien, Bo-Zhou Xie, Yu-An Chen, Cheng-Shiuan Tsai, Shao-Jyun Doong, Jun-Chen Hung.
MIT Sloan Sports Analytics Conference 2025, Research Papers Competition Finalist.

Invited Talks

Test-Time Decision Making: From Reinforcement Learning to Large Language Models
Institute of Statistics and Data Science, National Tsing Hua University — May 12, 2026

Honors & Awards

Excellent Internship Award, Taiwan Semiconductor Manufacturing Company (TSMC) — Oct 2024

Master Thesis Award, Taiwanese Association for Artificial Intelligence — Nov 2023

Outstanding Doctoral Student Scholarship, Ministry of Science and Technology — Sep 2023